

RetinaSight AI

Offline, on-device diabetic retinopathy screening

India has **125 million diabetics** who each need an annual retina check, and **one retina specialist per 1.26 million people**. Cameras already exist in vision centres. Images get taken — then queue for a human grader for weeks, and the patient has gone home.

We move the referral decision to the point of care: graded on the phone itself, offline, and spoken aloud in the patient's own language.

0.9324

QUADRATIC KAPPA

98.2%

REFERABLE SENSITIVITY

101 ms

ON-DEVICE LATENCY

11

LANGUAGES, WITH VOICE

0

SERVERS IN THE PATH

Final round pitch · Medical concept · Technical stack · Business model | Every figure measured or cited. Screening and triage support — not a diagnosis, no regulatory clearance.

The medicine

Why the eye is where diabetes shows itself first — and why this is a screening problem rather than a treatment problem.

The retina is a window onto the blood supply

Diabetes is a disease of **small blood vessels**. Persistently high blood sugar weakens capillary walls throughout the body. The retina matters for one reason that is easy to remember:

The retina is the only place in the human body where you can look directly at blood vessels without cutting anything open.

Shine light through the pupil, photograph what returns, and you are looking at live microvasculature. That is why the eye is diabetes' early-warning system for the whole body.

The five grades — what is physically happening

This is the **ICDR scale**, the international clinical standard. Our model outputs exactly these five.

Grade	What is happening in the tissue		Visible sign
0	No DR	Vessels intact	Clean field, sharp vessels
1	Mild	Capillary walls balloon where weakened	Microaneurysms — tiny red dots
2	Moderate	Vessels leak blood and fats	Dot/blot haemorrhages, hard exudates, cotton-wool spots
3	Severe	Vessels block; retina starves (4-2-1 rule)	Haemorrhage in 4 quadrants, venous beading, IRMA
4	Proliferative	Starved retina grows fragile new vessels	Neovascularisation , bleed risk, detachment

The clinically cruel fact

Vision stays normal until it suddenly isn't. The macula can be soaked in exudate and the patient still reads fine. There is no pain. By the time they notice blurring they are often Grade 3–4 and the damage is permanent. Caught early, DR is very treatable — laser, anti-VEGF, sugar control.

So the entire clinical problem collapses to one question: **who needs an ophthalmologist, before they have symptoms?** That is a triage question, not a diagnosis question. It is the only question we answer.

The Indian bottleneck

Fact	Value	Source
DR prevalence among Indian diabetics	12.5%	IJO National Survey 2015–19
Sight-threatening DR	4%	same
Retina specialists	1 per 1.26 M	PMC, DR screening in India
Diabetics needing annual screening	~125 M	Remidio / PR Newswire

The bottleneck is **not cameras and not patients**. Both exist. The referral decision is the thing that arrives too late — and that is the only thing we move.

The constraint we state before anyone asks

A smartphone camera **cannot** photograph a retina — it needs optics through the pupil. RetinaSight is the grading and workflow layer. Capture requires a fundus adapter or an existing clinic camera. Our capture screen says exactly this, in all eleven languages.

The technology

A convolutional neural network trained with an ordinal-aware loss, exported to ONNX, running on the phone's neural processing unit.

The pipeline — all seven stages on the phone

```
fundus photograph
↓
[1] quality gate      → REJECT (retake) or PASS
[2] preprocessing    → 456×456 normalised tensor
[3] ONNX model (NNAPI) → logits(1,5) · cam(1,5,15,15) · grade(1)
[4] derived values    → grade · confidence · expected grade · heat map
[5] referral decision → refer / do not refer
[6] urgency + advice  → what to do, in 11 languages
[7] speech + local record

The photograph never leaves the device.
```

Why a CNN, and why 456 pixels

A retinal lesion is a **local texture pattern**. A CNN applies the same learned filter at every image position, so a microaneurysm is detected wherever it sits — that property is translation invariance, and it is exactly what this problem needs.

Standard vision models use 224×224. We use **456×456**, because a microaneurysm is a handful of pixels; downsample to 224 and it is literally averaged out of existence — and it is the entire difference between "come back next year" and "you have diabetic eye disease".

Architecture and training — real configuration

Model		Training	
Architecture	EfficientNet-B0	Optimiser	AdamW, 2e-4, wd 1e-4
Input	456×456, NCHW	Scheduler	CosineAnnealing
Parameters	~5.3 M	Precision	AMP mixed fp16/32
Export	ONNX opset 17, 16 MB	Batch / dropout	8 / 0.4
Init	ImageNet pre-trained	Epochs	25 of 30, best @ 15
PyTorch↔ONNX parity	1.09e-4	Dataset	APTOS 2019, 3,662 imgs

Our core technical novelty — the ordinal-aware loss

Standard cross-entropy treats the five grades as five *unrelated* categories, so predicting Grade 0 for a Grade 4 patient costs exactly the same as predicting Grade 3. That is clinically absurd — one of those errors sends a nearly-blind patient home.

```
loss = weighted CrossEntropy(label_smoothing 0.05)
      + 0.5 × MSE(expected_grade, true_grade)

expected_grade =  $\sum_i i \times \text{softmax}(\text{logits})_i$ 
```

The second term is a **squared** distance, deliberately mirroring the quadratic weighting inside the metric. Being wrong by three grades is penalised **nine times** more than being wrong by one. This single choice is why our kappa is 0.93 rather than roughly 0.85.

Measured accuracy — held-out split, n = 546

0.9324

QUADRATIC KAPPA

83.3%

EXACT ACCURACY

0.8398

WEIGHTED F1

98.2%

REFERABLE SENS.

Grade	Precision	Recall	F1	Support
0 · No DR	1.0000	0.9741	0.9869	270
1 · Mild	0.5938	0.6909	0.6387	55
2 · Moderate	0.7836	0.7047	0.7420	149
3 · Severe	0.4043	0.6786	0.5067	28
4 · Proliferative	0.7895	0.6818	0.7317	44
Macro average	0.7142	0.7460	0.7212	546
Weighted average	0.8525	0.8333	0.8398	546

Read the band, not the diagonal. Almost every error is off by exactly one grade. No healthy eye was ever called severe or proliferative; no proliferative eye was ever called healthy or mild. *The catastrophic errors do not occur.*

The referral decision — separate from the grade

The grade is what the model says. The **referral is what the patient must do**, and it is a different decision. We refer at an expected grade of **1.15**, deliberately below the rounding point of 1.5.

Threshold	Sensitivity	Specificity	Missed	False alarms
1.50 — the rounding point	91.4%	94.8%	19	17
1.15 — what we ship	98.2%	92.9%	4	23

Fifteen fewer patients missed, for six more unnecessary clinic visits. Those errors are not symmetric and we refuse to pretend they are. A missed proliferative patient goes blind; a false alarm wastes an afternoon.

On-device stack and measured performance

Stack		Measured — vivo iQOO 15	
Language / UI	Kotlin · Jetpack Compose	Latency p50	101.38 ms
Camera	CameraX	Latency p90	109.65 ms
Inference	ONNX Runtime for Android	Latency p99	116.12 ms
Acceleration	NNAPI → NPU/GPU, CPU fallback	Energy / screening	0.33–0.37 mWh
Speech	On-device TTS, 11 languages	Quality-gate checks	6

What we say before a judge finds it

- **This is internal validation** — one dataset, one set of cameras. Not clinical performance. External validation on IDRiD is next; the harness is written and verified, only the data is missing.
- **Grade 3 precision is 0.40.** We over-call severe — which for a screening tool errs in the safe direction, and only 28 severe cases exist in validation.
- **No regulatory clearance.** Our own model metadata says `clinically_validated: false`.
- **We do not claim to beat FDA-cleared systems.** EyeArt's 96%/88% comes from a prospective multicentre trial against dilated stereo grading — a far higher evidence bar than a dataset split. Our claim is the engineering: on-device, offline, 101 ms.

The business

B2B and B2G. The diabetic patient is the beneficiary; the health system is the customer.

Market

Market	Value	Source
Global AI-DR screening, 2024	US\$0.40 B	DataM Intelligence
Global, 2033 projected	US\$2.22 B	DataM Intelligence
Global CAGR	~21%	DataM Intelligence
US market, 2026 → 2035	\$231 M → \$1.31 B	Towards Healthcare

Growth drivers cited across sources: rising diabetes prevalence, reimbursement pathways, and **shortage of ophthalmologists and access gaps** — precisely the gap we address.

Revenue model — ₹40 per completed AI screening

A software fee, not the cost of the eye examination. Priced this way because our **marginal cost per screening is effectively zero** — inference runs on hardware the customer already owns. Every cloud competitor pays GPU cost per image, forever. That is a structural gross-margin advantage, not a discount.

Share of the 125 M annual screenings	Screenings / year	Revenue / year
0.1%	125,000	₹50 lakh
1%	1.25 million	₹5 crore
5%	6.25 million	₹25 crore
10%	12.5 million	₹50 crore

Pricing is our hypothesis, not a market-report figure. Grounding: Kerala's public DR screening pilot achieved ₹22,000 per QALY — already highly cost-effective against India's GNI per capita (Eye / Nature). We lower its marginal cost further.

Customers

Segment	Why they buy
Government programmes Ayushman Bharat, NPCB	Screen at PHC scale without hiring ophthalmologists
NGO networks Aravind, LVPEI, Sightsavers	More camps per rupee of donor money
Hospital & diabetes chains	In-house triage; retain the referral inside the network
Fundus camera OEMs	White-label AI that makes their hardware more saleable

Competitors, and the wedge

Player	Status	The gap we exploit
Remidio Medios India	First CDSCO-approved ophthalmic AI in India;	Locked to Remidio's own camera. You buy their hardware or you cannot use their AI

	offline-capable	
Eyenuk EyeArt US, FDA cleared	96% sens / 88% spec (mild DR), n=915 prospective trial	Cloud-dependent, US-priced, needs connectivity
IDx-DR US, first autonomous FDA clearance	87% sens / 90% spec	Bound to specific cameras; US reimbursement model
Google ARDA	Research and deployments	Cloud inference; no offline path

Five things only we can say

- **Hardware-agnostic.** Any fundus image source, any Android phone. India's ~25,000 vision centres already own cameras from many vendors and do not want to re-buy. *This is the commercial wedge.*
- **Genuinely offline.** No server in the diagnostic path — demonstrable in airplane mode. Cloud competitors degrade to useless exactly where need is greatest.
- **Eleven Indian languages, spoken.** The difference between a referral being acted on and ignored.
- **Published field telemetry.** Milliseconds and milliwatts measured on the device. No competitor publishes this.
- **Open weights, Apache-2.0.** A state health department can audit the model before deploying it to a population.

Risks — stated honestly

Risk	Mitigation
No regulatory clearance — the gate for government deals	CDSCO pathway is the roadmap; this is our biggest real obstacle
Internal validation only	External validation on IDRiD is the immediate next step
Incumbent has approval and the Aravind relationship	We compete on hardware-agnosticism and languages, not on clearance
Liability for a missed case	Screening support with mandatory clinician confirmation; threshold biased toward over-referral

Go to market

1. NGO screening camps — field validation, real data, credibility · 2. State health department pilots — Kerala and Tamil Nadu have existing DR programmes to plug into · 3. OEM white-label — camera manufacturers bundle it, giving national reach without a direct sales force.

We are not replacing the ophthalmologist.
*We are deciding **who needs one** — while the patient is still in the chair.*

Sources: DataM Intelligence · Towards Healthcare · Ophthalmology Times (Remidio CDSCO) · Ophthalmology Science (EyeArt pivotal trial) · Indian Journal of Ophthalmology (National DR Survey 2015–19) · PMC (DR screening in India) · Eye / Nature (Kerala cost-effectiveness) · PR Newswire (Remidio India). All model metrics reproduced from the shipped ONNX artefact on a held-out split of APTOS 2019.